ERC Consolidator Grant 2027

Research proposal [Part B1]

A New Web: Bridging NFPs for Heterogeneous Societal Requirements

ISLANDS

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Proposal duration: 60 months

Abstract

The World Wide Web scaled to billions of users through architectural decisions that gave up stable references, versioning, and bidirectional links. Its downsides show where information has societal consequences: a fabricated audio clip of a party leader circulated, unverifiable, during Slovakia's 2023 pre-election silence; over half the URLs in UK parliamentary answers of 1997–2006 no longer resolved by 2007; a platform's EU transparency report was silently replaced with a corrected version six months after publication, through an updated hyperlink; and copyright courts from Munich to New York must reconstruct online attribution trails. *The Web's architecture hence deferred, rather than avoided, the cost of coordination, and society now pays it.*

ISLANDS extends today's Web with an architecture organized around *non-functional properties* (NFPs). *Islands* are bounded networked information spaces whose members choose the NFPs their context requires and provide them within each space, while islands are kept interoperable and backwards-compatible through today's Web's low-coordination architecture. With affected communities, we first establish which islands society requires, starting from legal citation and copyright, supply-chain carbon accounting, public-interest social media, signed research claims, and fixed industrial installations. We then derive the link classes these islands need, from *immutable, span-addressed versions* to references resolved through *confidential links*, and research how NFPs can translate at island borders. In addition to an open reference architecture and implementation that we validate with each island's community, we provide a *theory of chosen guarantees* predicting which NFP bundles an island of a given size and threat model can hold, at what coordination cost. Together, our architecture and theory will give society an extended Web whose guarantees are chosen by the communities that require them, and on which courts, parliaments, and markets can rely.

Cross-panel / interdisciplinary explanation

The proposal is computer-systems research and is submitted to **PE6 (Computer Science and Informatics)**. It spans four PE6 descriptors: PE6_10 (Web and information systems) as its core, PE6_2 (distributed systems), PE6_5 (cryptology, security, privacy; the confidential references and border checks of O3 and O4), and PE6_7 (artificial intelligence, intelligent systems, multi-agent systems; the agent layer of O4). We expect the scientific contribution to lie in the composition, which none of these sub-fields produces by itself.

Two validation domains and one objective also engage legal and governance scholarship (**SH2**, descriptor SH2_4, legal studies): we design the legal-citation island with legal practice and the social-media island against the data-access and transparency duties of the Digital Services Act, the interoperability duties of the Digital Markets Act, and the forthcoming Digital Fairness Act, and the stakeholder analysis (O1, O5) builds on legal scholarship on architecture as regulation, on RFC 9518's centralization analysis, and on the PI's published work on DSA auditing and transparency regulation. The same analysis covers the business models, costs, and incentives that sustain each island, so that the proposal is interdisciplinary in its assessment of the islands while its methods and deliverables remain in computer science. External SH2 expertise, including on the economics of institutions, would sharpen the assessment of O1 and O5; the methods and deliverables are PE6, and SH2 is flagged as the only secondary panel.

Section a: Part I of the Scientific Proposal

1. The problem and the idea

The World Wide Web's original design favored simplicity, decentralization, and low coordination overhead, which enabled explosive adoption but sacrificed stable references, built-in versioning, and native bidirectional linking [1, 2]. This was a deliberate choice: Berners-Lee's 1989 proposal required that the system "allow existing systems to be linked together without requiring any central control or coordination" [3]. When Fielding and Taylor defined the Representational State Transfer (REST) architectural style, they clarified the resulting architectural constraints: scalable component interactions, uniform interfaces, independently deployable components, and intermediaries that can be inserted anywhere [4]. These constraints enabled the Web's scaling up to billions of users and devices within three decades. We argue that the Web's trade-offs have, however, become too rigid for today's use cases. Today, publishers require *traceable* dereferencing, courts and parliaments require *stable, specific* citations, supply chains require aggregation over *confidential* data, users should retain *sovereignty* over their data, and society requires *trusted* content and *accountable* authorship. These needs should be satisfied *architecturally*, but in a way that is backwards-compatible with today's Web.

While we have kept the Web's architecture fundamentally static, *platforms* have responded: Accounts, persistent storage, search, reputation, and payment do not exist between independent Web sites, but platforms provide them inside their "walled gardens" [5] into which they draw the Web's content, users, and interactions [6], and they extend into the rest of the Web through APIs and plug-ins on their own terms [7]. Users then rebuild continuity through archives and off-platform link hubs [8]. Berners-Lee warned already in 2010 that social networks were walling off information their users posted [9], and RFC 9518 describes how functions built on open protocols *still centralize* through network effects and switching costs [10]. The consequences are borne by society: Engagement-optimized ranking amplifies some political content over other content [11], one of several systemic risks that the EU's Digital Services Act (DSA) attempts to address, with limited success [12, 13, 14]. The data that platforms do release may change silently: in August 2026, six months after publication, TikTok repointed the download link of its DSA transparency report to a corrected version without announcement; the correction lowered the reported terrorist-content notices from 77,729 to 43,119 [15]. Press publishers face the same opacity: in October 2026, the Paris Court of Appeal found that Meta had supplied only fragmentary, opaque, and unverifiable data on its use of their content and upheld an order that Meta negotiate neighboring-rights payments with French media in good faith [16]. And generative AI assistants, running on essentially the same information system architecture that was created for human researchers in the 1990s, now repeat and amplify the pattern: chatbots and agentic browsers return answers in place of links and so control access to the Web's content [17], while, according to the W3C's Technical Architecture Group (TAG), a user agent acting on its user's behalf owes "meaningful visibility into the sources it relied on" [18]. And Reddit, a Very Large Online Platform (VLOP) under the DSA, announced in October 2026 that it is moving to a logged-in experience to protect its data from scraping [19] after AI answers cut its referrals by more than a million per week. This shows that, today, the properties that high-stakes communities require are available, but they are provided inside closed platforms and under these platforms' control: walled gardens instead of an open Web.

Society pays for these architectural omissions: Web references point to locations whose *content may change or vanish*, requiring workarounds; for instance, the UK government has created a redirect service in response to finding that more than half of the roughly 4,000 URLs cited in ministerial answers to Parliamentary Questions since 1997 no longer resolved [20]; one in five scholarly articles cites Web content that has since rotted [21], necessitating archival infrastructure such as Zenodo [22]. The Web *does not bind content to its author*: two days before Slovakia's 2023 election, a fabricated recording of a party leader discussing vote-rigging circulated with no provenance that voters could check [23]. The Web's architecture does *not support publishing verifiable aggregates over private data*, yet the EU now requires exactly this from core industries. From February 2027, every electric-vehicle and large industrial battery sold in the EU must carry a *digital battery passport* with its carbon footprint and supply-chain due diligence [24]. The Ecodesign for Sustainable Products Regulation (ESPR) extends such *digital product passports* to nearly all products on the single market, starting with steel in 2026 and textiles in 2027 [25, 26], and passports must stay available if their issuers disappear [25]. Web links *address whole resources*, not passages or versions of pages, while in law references need to be specific and stable; the EU therefore created persistent identifiers for judgments and legislation through ECLI and ELI [27, 28]. Because the Web *records no provenance*, content provenance is being moved into other places, from camera hardware [29] to EU law [30, 12]. Finally, the Web *affords copying content* instead of transcluding it [31], hence reuse by third parties leaves no trail; this ultimately leads to the Munich Regional Court holding in November 2025 that a language model's memorization of song lyrics infringes copyright [32], a similar question to what *The New York Times v. OpenAI* is currently litigating in the U.S.

[33]. *What all these cases have in common is that coordination cost that the Web avoided was deferred, not eliminated; society now pays for this architectural misalignment, one incident at a time.*

The Idea. ISLANDS proposes to *extend* today's Web's architecture with specific architectural guarantees. Communities that require these guarantees can then select their *own property configuration*, i.e., a set of desired *non-functional properties* (NFPs) that apply to their part of the Web. One example of such a community is the *copyright community*, i.e., all stakeholders with a pronounced interest in having content rights preserved by the architecture, from authors and publishers to courts. Another is an industry that needs to account for carbon emissions along its supply chain, which requires the architectural preservation of confidentiality of *individual business data* while *aggregates* are published and verified. To enable this extension, we define *islands* as bounded networked information spaces that cater to such communities with explicitly selected NFP bundles. An island's members share stakes high enough to justify the cost of architectural coordination, and they settle it through an ex-ante selected architectural trade-off rather than ex-post patching, fragmentation through platformization, or litigation. With ISLANDS, *The New York Times*, for instance, could choose to publish its articles into an island whose NFPs bind every quotation to its source through traceable linking and that constrain text and data mining [34, 35]. Readers and AI agents that accept these terms enter the island, where terms and attribution travel with every quotation to participating sites and agents, and reuse becomes verifiable. Copying a quotation out of the island remains possible, since islands do not restrict reading or copying, but the copy leaves the island's guarantees behind: it no longer verifies against its source, and readers, courts, and agents can thereby tell it apart from an attributed quotation. With today's non-architectural alternative, i.e., platformization and paywalling, control is confined to individual platforms, and once an article has been read by an AI agent or copied by a user, its quotations and copies leave without terms or provenance.

Each island selects the NFPs its participants require and provides them jointly within its scope, while islands interoperate with one another and with the open Web through the Web's existing low-coordination mechanisms that permit planetary scaling. *Within* an island, references can for instance be span-level, version-pinned, bidirectional, and verifiable, similar to how Xanadu specified them in the 1980s [36]. *Across* the boundary, they degrade to ordinary Web links, so that every island remains backwards-compatible with today's Web. The specific NFPs that can be selected within an island are flexible: In ISLANDS, we foresee *confidentiality-preserving references* that resolve to a value, such as an aggregate, that members can verify was correctly derived from encrypted inputs; this yields an *architectural integration* of zero-knowledge proofs [37] and secure multi-party computation (SMPC) [38]. NFPs also include *purpose binding*, which attaches the terms of a disclosure (e.g., recipient, purpose, validity window) to the reference, so that a reference is re-resolved when its terms expire [39]. *Attribution* binds a reference to an accountable author or to an eligible but unidentified one through the European Digital Identity Wallet (EDIW) [40, 41]. *Verifiable credentials* let a reference carry an issuer's attested claim about its source, e.g., that a filing was signed by a member of the bar or that an emissions figure was checked by an accredited verifier [42]. And *capability discovery* architecturally extends the Web's *Hypermedia as the Engine of Application State* (HATEOAS) principle [2] to cater to artificial agents, as recently demonstrated in the multi-agent systems field [43, 44]. Finally, an island of fixed, high-speed installations may bind its interfaces once, at design time, in a binary encoding, much as a linker resolves a program's symbols when it builds the program [45]. This forgoes the Web's extensibility and evolvability in exchange for performance, while it follows the same ISLANDS principle of an explicitly selected NFP bundle. Our unifying scientific contribution is hence a *theory of chosen guarantees*, which is a formal account of *which NFP bundles an island can provide in principle, at which coordination cost, under which threat model, and for which stakeholders*.

Figure 1 shows five proposed islands, embedded in the Web. Each modifies the Web's basic contract to support the NFPs that its community requires. As an example, *Island L* lets a court or author cite a passage (through *span addressing*) that resolves to exactly the version relied on (through *persistence*). It also leaves a trail of reuse (through *backlinks*) that a copyright dispute, or any scheme that remunerates reuse, can follow, and it makes terms such as a rightsholder's reservation against text and data mining [34, 35] verifiable in the reference (through *attribution* and *credentials*). It does not, however, establish that a cited source is correct, and it records no reuse that happens outside the island. *Island S* lets suppliers report confidential emission figures that can be checked without decryption (through *confidential references*); *Island M* lets readers see who stands behind an account and how a post was edited or deleted (through *persistence*, *attribution*, and *credentials*), and lets researchers obtain exposure statistics (through *confidential references*); *Island N* keeps signed, content-hashed research claims, as published in a nanopublication network [46, 47], citable and attributed (through *span addressing, persistence, and attribution)*. *Island F*, finally, lets fixed installations interact over binary interfaces at low latency while giving up the Web's evolvability (through *static binding*).

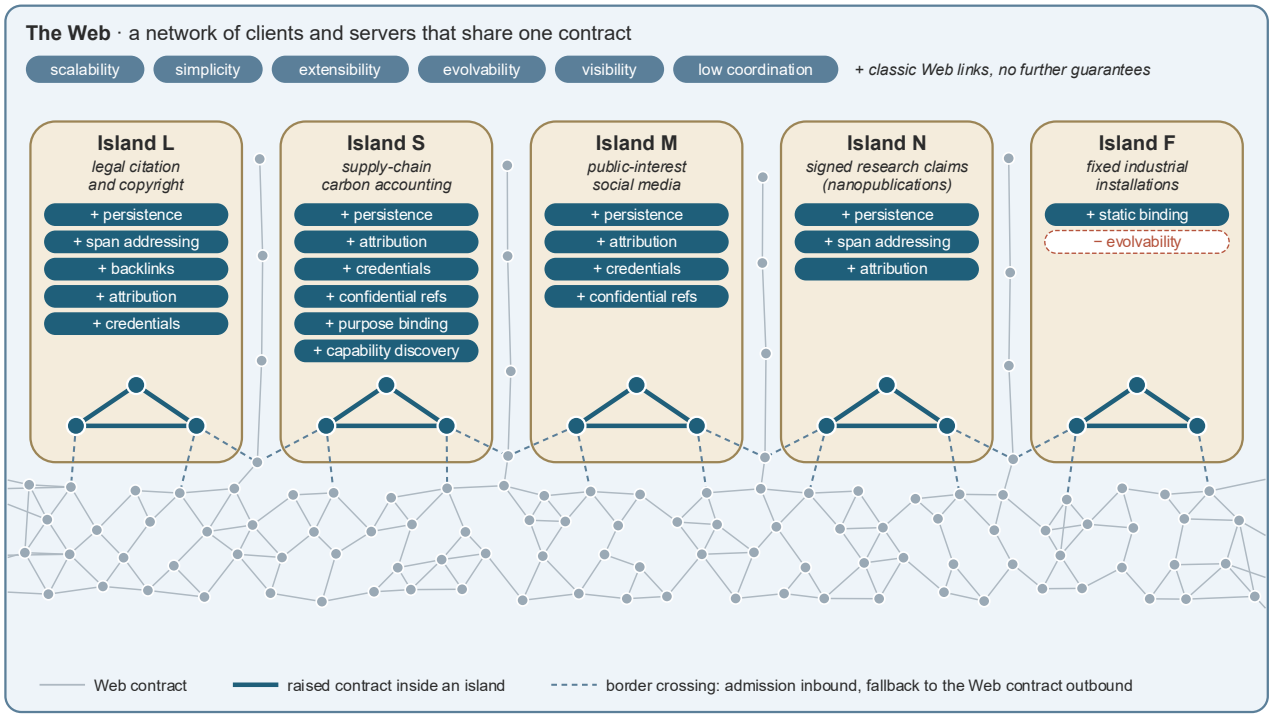


Figure 1. Five example islands within the Web. Outside any island, references are classic Web links without further guarantees (thin links). Within islands, members raise the contract by adding NFPs (filled pills), so that links inside the island carry stronger guarantees (thick links). Links that cross an island's border pass admission inbound and fall back to the Web contract outbound (dashed).

Possible and Required Today. Building these properties into the architecture did not pay off in 1989, but it does today. First, the coordination required by many of the important NFPs has become far less costly. Content addressing [48], *verifiable credentials*, a W3C Recommendation since 2025 [42], practical *secure multi-party computation* [49], and *decentralized version control* (through Git), are standardized and deployed at scale, and every EU Member State must offer an EDIW since 24 December 2026 [41]. Second, demand for the same guarantees now arrives from many directions at once. The DSA requires attribution and research access [12], the Data Act portability [50], the AI Act respect for machine-readable reservations against text and data mining [35], and the battery passport and CBAM verifiable supply-chain data [24, 51]. Today, every platform implements each of these obligations separately, and every regulator checks them separately: N platforms and M obligations yield N × M integrations. An architecture specifies each obligation once, as an NFP bundle, and lets each platform implement it once, which turns N × M into N + M. Third, platform-level solutions are failing visibly today. A platform can, as shown above, withdraw or silently change what it offers, and machines, which now produce most Web traffic [52], cannot check a platform's promises; machine “negotiation” relies on crude mechanisms such as robots.txt [53] or recent patches such as the IETF's vocabulary for AI usage preferences [54]. Finally, the protocols that will carry these guarantees are being standardized now, in the W3C groups on the social Web, storage, credentials, and agents and in agent protocols such as MCP [55] and x402 [56].

2. State of knowledge and research gap

Hypertext research traditionally assumed rich, span-anchored links (e.g., [36, 57]), and its agenda included versioning and computed structure (e.g., [1]). A link would *persist*, and it would point at a *specific range of bytes* rather than a full page. Modifying a link would create a new version instead of replacing the link. Open-hypermedia systems implemented *bidirectional* links and *referential integrity* at workgroup scale [58, 59]; link rot had thus been addressed even before the Web made it endemic. The Web dropped these guarantees to scale [3, 4], and this was a sensible and pragmatic choice at the time.

Since the mid-1990s, when the *Internet Archive* began to archive the Web [60], remedial infrastructure has kept this architecture and compensated for its drawbacks, such as link rot and misattribution. This infrastructure includes persistent identifier overlays, Web archives, Memento [61], Perma.cc [62], and robust hyperlinks [63], with permanent links such as those of the *Internet Archive's Wayback Machine* or, in the research space, *Zenodo* [22], since link rot undermines reproducibility [21]. Such compensation did not touch the architectural principles of the Web, since changing them might fragment the "one web" [64]. W3C specifications also identify passages (Web Annotation selectors [65], text fragments [66]) and deliver opt-in backlinks (Webmention [67], Linked Data Notifications [68], and ActivityPub [69]); however, none of them pins a version or verifies what it points to. Dedicated services assemble specific NFP profiles as well, e.g., persistence

via *Perma.cc* or *Zenodo*, provenance via *C2PA* [29], and identity and notarization via *DOIs* [70] or *blockchains* [71]. *Solid* [72] brings *data sovereignty* as well as *privacy and access control*, since users keep their data in personal stores of their choice and control access. Remuneration for reuse followed a similar path. Xanadu already tied *transclusion* to micropayments, so that each delivered fragment would earn its owner a royalty, which Nelson later recast as *transcopyright* [73]. Micropayment schemes for the Web [74] did not take hold, but they now return for machine readers: the status code 402 "Payment Required", reserved in HTTP for decades [75], is being used to charge AI crawlers per fetch [76] and agents per request [56]. None of these schemes, however, rests on a record of use that is built into the architecture and that both sides can verify. Still, they are evidence of the demand for chosen guarantees that extend beyond those of the basic Web architecture. As a community, we are thus aware of these issues and have built remedies. However, each is a stand-alone solution that provides one fixed profile; their packaging in NFP bundles, as proposed in ISLANDS, is novel. The reasons why there is no architected approach yet include that the technologies were not available and that the pressure mounted slowly; AI agents mark a turning point. Further, an architected approach is hard since it requires consolidating fields that have so far advanced separately, including hypertext, distributed systems, cryptography, multi-agent systems, and law.

The methods that make such a shift possible today (Section 1) were not available in the early days of the Web: *Content-addressed* and *versioned storage* (Git; IPFS [48]; Software Heritage identifiers [77]) and *provenance attestation* [29, 78] now supply, at Web scale, possibilities that Xanadu [36] and other architectural designs lacked. They are, however, technical storage and signing primitives without span semantics, a link layer, or a governance model. ISLANDS takes from Xanadu its traceability of reuse, not control over reading: a quoted passage stays linked to its source, but nothing restricts copying or enforces terms on readers' devices; however, readers and devices will know when they consume content outside of a desired island. *Nanopublications* [46] form a deployed, federated network of immutable, signed semantic statements with self-verifying identifiers. They do not, however, offer span addressing or transclusion. With respect to *confidentiality*, communities need results computed over data that no single member may disclose: SMPC has run in production since 2008 [38] and general-purpose protocols and implementations such as SPDZ [79] and MP-SPDZ [49] make computation practical. Our own PEP-LCA protocol computes joint supply-chain assessments across up to 50,000 participants without a trusted third party [80], and the disclosure of shares can be controlled through Solid [81]. Again, each of these deployments is an individual application, and none exposes its result as a reference that others can cite and re-resolve. *Verifiable credentials* allow making statements about who stands behind published data on the Web (cf. [42, 82, 41]), but they are presented to one verifier within one session rather than being bound to the references that a claim concerns. Finally, hypermedia multi-agent systems research argues for run-time *affordance* discovery [43, 83]. Current LLM-agent practice, in contrast, uses statically linked tool lists [55], and no existing design connects *capability discovery* to a verifiable reference layer.

Research Gap. Today, we do not know how different NFP selections work together in bundles. Each hypertext system and each remedy above fixes some guarantees by design, but they do not tell us which bundles are technically feasible in the first place, at what coordination cost, and against which threat model. It is open, for instance, whether *confidentiality-preserving references* can be combined with *bidirectional linking* without the backlinks revealing what the references hide, or whether *attribution* and *anonymity* can coexist in one island. This is a major gap, and we propose addressing it at the architectural level. Closing the gap requires an understanding of each of these domains, spanning distributed computing, decentralization, and the societal requirements behind them. For practical reasons, it must also be done in a backwards-compatible way that reuses the Web's architecture as deployed today and that enhances its NFPs only in bounded subsystems. Finally, it needs a theory that predicts which NFP bundles fit a given deployment context and community.

3. Objectives

ISLANDS asks: *which architectural guarantees can a Web-embedded island let communities choose, and at what cost?* Its goal is to answer this question for the communities that need such guarantees and to demonstrate the answer in practice. This closes the gap stated above, so that society no longer pays piecemeal for guarantees that the architecture lacks. It involves:

- **O1: Societally required islands.** We establish which *islands*, i.e., which bundles of NFPs, society requires today and in the foreseeable future. We start from the five drafts of Figure 1 and consider further possible islands while involving appropriate user communities.
- **O2: Theory of chosen guarantees.** We isolate the *link classes* that the required islands need and develop a formal model that relates them to coordination costs, island size, threat models, and stakeholder incentives. We foresee this taxonomy to include a ladder of link classes (*Classic Jumplinks*, *Archived*, *Trusted*, *Verified*, and *Cryptographic Links*, in increasing strength and cost) and at least five cross-cutting

axes (attribution, trust, usage, modality, *link time*). For each class, we prove what it minimally requires, and we derive decision procedures that an island designer can apply.

- **O3: Border translation.** We determine how NFPs, and the concepts that support them (versions, identities, credentials, usage terms, encodings), are translated when information enters or leaves an island. This includes which references are upgraded at admission, which are downgraded on exit, and how the loss and gain of guarantees happens in a way that is well-understood by human and artificial users.
- **O4: Island technologies.** We build the technologies that the required islands need. This will, minimally, include a reference and *transclusion* layer over content-addressed storage; confidential references that turn SMPC-based resolution into an addressing primitive; and an agent layer with state-dependent *capability discovery* and design-time *static links*.
- **O5: Validation and incentives.** We instantiate and evaluate the implemented islands with their communities and assess the technical artifacts of each island from business, legal, and economic perspectives: which business models can sustain the island, whether its guarantees meet the law of its domain, and how it shifts costs and incentives.

4. Research strategy and team

Our project's work plan follows the objectives above (see Part II): After establishing the islands that society requires, we define link classes and start the theory of chosen guarantees. In parallel, we derive the border checks into and out of islands, considering the requirements of both human and machine users. From the beginning, we involve the islands' target communities and deploy the created island technologies with them for validation. We do not start this project from an empty design, but from communities that we are already working with and from technical building blocks that are proven individually, including in systems that our team has built. ISLANDS adds on top of this that communities can select NFP bundles *as an architectural addition to the Web*. All project software and publications will be open, and created specifications are shared with the W3C groups that maintain the individual technologies and protocols that ISLANDS builds on.

Team. Answering ISLANDS' research question needs a team that covers Web architecture, human-computer interaction, multi-agent systems, and law, with additional expertise from cybersecurity and economics for individual aspects of the project. The PI integrates the four main research lines in his research and team, and has demonstrated expertise in the two secondary lines. Each of his research lines has so far addressed one part of the problem; with ISLANDS, they are consolidated into the *theory of chosen guarantees* and the *architectural principles* for islands' link classes and border checks. The PI stepped down as Dean in 2026 to pursue this consolidation at more than 40% commitment; this consolidation is required because an island's guarantees are at once *architectural*, experienced by *people* and *agents*, and *legally* consequential. The PI has experience running large-scale research as well as innovation projects, his recognition in the community is demonstrated for instance through his role as the General Co-Chair of the 2027 Solid Symposium and the 2028 ACM CS&Law Conference, and two members of the PI's current research group would join the project and bring in their expertise and built systems, mitigating recruitment risks at project start.

5. Ambition, risk, gain

ISLANDS tests whether a coordination cost that was judged prohibitive when the Web's design was established [3, 4] is sustainable at *island* scale with today's technologies and approaches. It also tests individual aspects of this problem, such as whether cryptographic reference resolution can reach interactive latency in a realistic distributed system, and whether actors with divergent incentives will adopt shared guarantees. Finally, it tests whether, for the communities that need them, guarantees built into the architecture cost society less than today's platform-level fixes. These are risky hypotheses. If they hold, ISLANDS will deliver a validated architecture for the parts of the information space where the Web's founding trade-off now fails at measurable cost, with large implications on parliaments, lawmakers and courts, scientific discourse as well as public discourse on social media, and for regulated confidential industrial data. ISLANDS furthermore considers autonomous agents acting on the Web's hypermedia fabric from the beginning, building on our work on hypermedia multi-agent systems [43, 83, 44]. If our hypotheses fail, our theory of chosen guarantees still delimits which guarantees are attainable, for whom, and at what cost.

The alternative is to continue as today, with piecemeal fixes whose costs society absorbs one incident at a time. Where this might lead is sketched, in exaggerated form, by the *Dead Internet Theory*, which holds that most online content and activity is already produced by bots [84]. We do not endorse the theory fully, but its premise is moving closer: automated traffic surpassed human traffic on the Web in 2024 [52], and language models degrade when trained on content that models generated [85]. A Web that cannot state who stands behind a document, which version a claim rests on, or whether a text was generated, becomes a space in which neither people nor machines can rely on what they read. Courts, parliaments, and markets would then have to rebuild

trust outside the Web, at far greater cost than the coordination the Web avoided. ISLANDS offers the architectural alternative: guarantees that communities choose and that hold where they are needed.

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Section b: Curriculum Vitae and Track Record

1. Personal details

Prof. Dr. Simon Mayer; born 16 July 1987; Austrian. ORCID 0000-0001-6367-3454.

Education. Dr.sc. ETH in Computer Science, ETH Zurich (defense 12 September 2014; 12.3 years before 1 January 2027, within the CoG window of 5–15 years). M.Sc. ETH in Computer Science (2010); B.A. in Economics and Business Administration, University of Zurich (2014); MAS in Secondary and Higher Education, ETH Zurich (2014).

Current position. Since 2018: Full Professor of Computer Science (Interaction- and Communication-based Systems), University of St. Gallen (USG), one of the four founding professors of its School of Computer Science; Director, Institute of Computer Science. Since 2020: Board Member, Research Center for Information Law, USG.

Previous positions. 2024–02/2026: Dean, School of Computer Science, USG (2022–2024 Academic Director; 2018–2024 Managing Director, Institute of Computer Science). 2023: Visiting Professor, University of Bologna. 2018–2020: Senior Scientist, Distributed Systems Group, ETH Zurich. 2017–2018: Area Manager "Cognitive Products", Pro2Future GmbH, and Project Scientist, Graz University of Technology. 2014–2017: Research Scientist, Senior Researcher, and Senior Key Expert, Siemens Corporate Technology, Berkeley, USA. 2015–2017: Visiting Lecturer, UC Berkeley and ETH Zurich. 2013: Visiting Researcher, Ghent University. 2012: Visiting Researcher, MIT.

2. Research achievements and peer recognition

Research achievements (most recent first within each line).

1. Bekavac, **Mayer**. *Broken Links and Fading Stories: How Dominant Online Platforms Subvert the Hypermedia Foundation of the Web.* ACM WebSci 2026.
2. Bekavac, **Mayer**. *Platforms' Research API Data Access: What Users See vs. What Researchers Can Retrieve.* ACM FAccT 2026; and Bekavac, Strecker-Bischoff, Garcia, **Mayer**, Tamò-Larrieux. *Scrutinizing Systemic Risks in Personalized Recommender Systems Through Sock-Puppet Auditing of VLOPs.* ACM Transactions on Recommender Systems 5(1), 2026.
3. Grau, Maftun, Garcia, **Mayer**. *Solid and Secure Multi-Party Computation for a Circular Economy.* Solid Symposium 2026, building on Grau's MSc thesis (2025, supervised by the PI with Garcia, Mitrokotsa, and Cheng).
4. Guitton, Tamò-Larrieux, **Mayer**. *AI and Law: How Automation Is Changing the Law.* Routledge, 2025.
5. Tamò-Larrieux, Guitton, **Mayer**. *A Computational Science Perspective on the Legal System.* Nature Computational Science, 2025.
6. Vachtsevanou, Ciortea, **Mayer**, Lemée. *Signifiers as a First-class Abstraction in Hypermedia Multi-Agent Systems.* AAMAS 2023; and Vachtsevanou et al. *Enabling BDI Agents to Reason on a Dynamic Action Repertoire in Hypermedia Environments.* AAMAS 2024.
7. **Mayer**, Fuchs, Brügger, Lian, Ciortea. *Improving Customer Decisions in Web-based E-commerce through Guerrilla Modding.* Nature Machine Intelligence 3(12), 2021.
8. Tamò-Larrieux, **Mayer**, Zihlmann. *Not Hardcoding but Softcoding Data Protection.* Technology and Regulation, 2021.
9. Ciortea, **Mayer**, Gandon, Boissier, Ricci, Zimmermann. *A Decade in Hindsight: The Missing Bridge Between Multi-Agent Systems and the World Wide Web.* AAMAS 2019.
10. **W3C Web of Things Thing Description.** The PI's doctoral and Siemens research contributed to the basis of the W3C Thing Description, which is deployed in industrial products (e.g., Siemens Desigo building management).

Peer recognition. General Chair, Solid Symposium 2027 and ACM Symposium on Computer Science and Law (CS&Law) 2028. Member of the European oversight board on Very Large Online Platform auditing. Associate Editor, PACM IMWUT (since 2021); Editorial Board, IEEE Pervasive Computing (since 2019); Executive Editor, Wiley Internet Technology Letters (2017–2023); Special Issue Editor, ACM TOIT (2023). Co-organizer, Dagstuhl Seminars 21072 and 23081. Two CHI 2025 Honorable Mentions (adaptive industrial robots; responsible ubiquitous personalization, DIS 2025). Finalist, Best Teaching Award, USG (2024).

Narrative. The achievements document three lines of work that ISLANDS brings together. In Web architecture, the PI's work on the Web of Things contributed to a W3C standard that is deployed in industrial

products (10), and his work on hypermedia multi-agent systems introduced signifiers as a run-time abstraction through which agents discover what they can do (6, 9); the agent layer of ISLANDS and its route to standardization build on both. In law and technology, the PI has worked with legal scholars on regulation expressed as technical architecture and on how automation changes the law (4, 5, 8), and his group has audited what platforms disclose under the Digital Services Act (2); this work grounds Islands L and M and the stakeholder analysis. In decentralized data, the PI's group built the secure multi-party computation system over Solid pods on which the confidential references of ISLANDS build (3), and its recent study of how platforms undermine the Web's links states the problem that ISLANDS addresses (1). User-side re-engineering of Web experiences (7) showed early that communities can impose their own guarantees on top of platform defaults.

3. Additional information

Career path. After his doctorate, the PI spent three years in industrial research at Siemens Corporate Technology in Berkeley (2014–2017), where the Web of Things work moved into products. From 2022 to February 2026, he held academic management roles at USG, first as Academic Director and then as Dean of the School of Computer Science, which he built to 10 research groups, more than 120 employees, and about 300 students. He stepped down as Dean in 2026 to refocus on research. The PI has three children (born 2015, 2017, and 2019).

Supervision. Five postdocs mentored into faculty or senior positions (Gómez, Associate Professor, TU Braunschweig; Garcia, Tenure-track Assistant Professor, IT:U Linz; Ciortea, Assistant Professor, USG; Guitton, Swiss Re; Bektaş, Senior Researcher, KSSG). Three PhD graduates in 2024/25 (Ramanathan, Hostettler, Jha), one completing (Vachtsevanou), and six current, including Bekavac and Grau.

Funding. PI or co-PI of 4 SNSF, 2 Innosuisse, 3 EU, and more than 12 industry or foundation projects (3.9 MCHF allocated to the PI's group out of 13.6 MCHF in total), including SNSF HyperAgents (2019–2024), EU IntellIoT (2020–2024), Chips JU AFarEdge (2026–2029, lead of the architecture definition), Innosuisse Flagship WISER (2022–2026), SNSF CoCoDa (2025–2029), SNSF/Innosuisse *Access to Justice* (2026–2030), and TA-SWISS *Sprachmodelle und Kompetenzverschiebungen* (2025–2026).

Contributions to the research community. Active in W3C and IETF standardization; founding member of the W3C Autonomous Agents on the Web Community Group. Steering Committee, International Conference on the Internet of Things (since 2017). Co-organizer of the International Workshop on Hypermedia Multi-Agent Systems (2019–2020) and of the Web of Things workshop series (2012–2017). U.S. Patent US11331803B2 (mixed-reality robot programming, with Siemens). Co-founder of the Future Society Hub (FuSo) at USG, which connects computer science and law.

Application note: Island L and Open Terms Archive

Open Terms Archive (opentermsarchive.org) records every version of the terms of service and privacy policies of digital services "to enable democratic oversight", and the European Commission's DSA enforcement team uses it. Contributors declare, per document, a URL and CSS selectors; robots fetch each document several times a day, store the source as a snapshot, filter it into a textual version, and commit dated versions to public Git repositories, which collections publish as datasets and through a federated API. Open Terms Archive also informs the European Commission about changes in the terms it tracks. Terms published in Island L would make most of this machinery unnecessary. A provider would publish each version of its terms as an immutable, signed record and each change as a superseding record, so that polling robots, change detection, and the dated commits that a third party makes on the provider's behalf would no longer be needed. The Commission would no longer depend on a third party to learn of a change either: it could subscribe to the provider's terms through *Live/Subscription Links* and be notified as soon as the provider publishes a superseding version, which the provider has signed and cannot later deny. Declarations with selectors and filters, and their maintenance whenever a provider redesigns its pages, would also become unnecessary, since the island holds the text itself and not a rendered page. Authenticity would rest on the provider's signature, which a regulator or court can check, instead of on trust in the archivist's crawl. Span addressing would add what the archive lacks today: a verifiable reference to the exact clause of the exact version that a user accepted, which a consumer, regulator, or court can cite. Open Terms Archive would remain necessary for providers that do not join the island, and those that benefit most from opaque terms are the least likely to join. Its collections, analyses, and federated search would remain valuable in any case, as would its memos that explain to the Commission what a change means, but they would work on verified records instead of crawled copies.

Example: a silent correction of a DSA transparency report. An Open Terms Archive investigation by Matti Schneider shows the difference in practice. On 28 August 2026, six months after publication, TikTok repointed the download link of its DSA transparency report for July to December 2025 to a corrected archive, without a public announcement. For terrorist content notices (Category 10d), the number of notices received fell from

77,729 to 43,119 and the median time to take action from 6.82 to 2.43, in a unit the report does not state; the earlier figures had been computed from reports of consumer-related offences. The parent category total was not recalculated, so the corrected file no longer adds up. Open Terms Archive noticed the change only because the report was linked from a page it tracked for another collection, and its analysts had to download both archives and compare them by hand. In Island L, the platform would publish the report as an immutable version and the correction as a superseding version under its signature. Each figure would be span-addressable, so a study that had cited the 77,729 notices through a Verified Link would still resolve to exactly that value, and the island's backlink registry would show the study that a superseding version exists. This answers the question that the investigation leaves open under Implementing Regulation (EU) 2024/2835, which requires versions of transparency reports to be marked and kept accessible: whether one can trace which conclusions rest on obsolete data. Island L would neither prevent a platform from publishing wrong or inconsistent figures, nor make a correction visible if the platform publishes outside the island.